

Name: \_\_\_\_\_

Reg No: \_\_\_\_\_

Quiz #4

1. **Q,K,V matrices come from:**
  - a. Different random init
  - b. Input  $\times$  3 learned weights
  - c. Previous layer only
  - d. Fixed function
2. **Why divide by  $\sqrt{d_k}$ ?**
  - a. Normalize output
  - b. Prevent large dot products  $\rightarrow$  vanishing gradients
  - c. Get probabilities
  - d. Make symmetric
3. **Multi-head:  $d_k = d_v = d_{model}/h$  because:**
  - a. Equal params per head
  - b. Same cost as single-head
  - c. Concatenation matches  $d_{model}$
  - d. B and C
4. **Encoder attention mask:**
  - a. Causal
  - b. Padding only
  - c. Bidirectional (with padding mask)
  - d. Triangular
5. **FFN purpose:**
  - a. Add non-linearity
  - b. Reduce length
  - c. Residual connections
  - d. Normalize
6.  $d_{model} = 768$ ,  $h = 12$  **heads**,  $d_k = ?$ 
  - a. 768
  - b. 12
  - c. 64
  - d. 128
7. **Cross-attention: Query from \_\_\_\_\_, K,V from \_\_\_\_\_**
  - a. Enc; Dec
  - b. Dec; Enc
  - c. Prev Dec; Same Dec
  - d. Input; Output
8. **Residual connections:**
  - a. Before LayerNorm
  - b. Prevent vanishing gradients
  - c. Encoder only
  - d. Replace positional encoding
9. **After concat heads:**
  - a. Direct to FFN
  - b. Multiply by  $W^O$
  - c. Average heads
  - d. Normalize & sum
10. **Masked self-attention: token  $i$  attends to  $j$  where:**
  - a.  $j > i$
  - b.  $j \leq i$
  - c.  $j \neq i$
  - d. All  $j$
11. **BERT-base ( $d_{model} = 768$ ,  $h = 12$ ,  $L = 6$ ) parameters approx:**
  - a. 10M
  - b. 110M
  - c. 1B
  - d. 10B
12. **Sinusoidal encoding advantage:**
  - a. Extrapolate longer sequences
  - b. Fewer parameters
  - c. Easier optimization
  - d. A and B
13. **LayerNorm applied across:**
  - a. Batch
  - b. Features per token
  - c. Sequence length
  - d. After activation
14. **What does positional encoding add to word embeddings?**
  - a. Order information
  - b. Semantic meaning
  - c. Attention weights
  - d. Vocabulary size
15.  **$QK^T[i, j]$  represents:**
  - a. Attention weight  $i \rightarrow j$
  - b. Attention weight  $j \rightarrow i$
  - c. Raw dot product  $q_i \cdot k_j$
  - d. Softmax output
16. **Transformers need more data than RNNs because:**
  - a. More params, no sequential bias
  - b. Cannot handle variable length
  - c. ReLU activations
  - d. Weak gradients
17. **Without  $\frac{1}{\sqrt{d_k}}$  scaling, softmax becomes:**
  - a. Uniform
  - b. Near one-hot
  - c. Unchanged
  - d. Zero
18. **Trained on length 512, tested on 1024. Result?**
  - a. Works perfectly
  - b. Crashes
  - c. Degrades beyond 512
  - d. Depends on encoding type
19. **During decoder training, what is Teacher Forcing?**
  - a. Using model's own predictions as input
  - b. Feeding true previous tokens as input
  - c. Masking future tokens
  - d. Scaling the dot product
20. **During inference, the decoder generates tokens until \_\_\_\_\_**

Each question carries 1 mark. No negative marking.

Score: \_\_\_\_\_/20

**Answer Key (For Instructor Use Only)**

1. B
2. B
3. D
4. C
9. B
10. B
11. B
12. D
17. B
18. D
19. B
20. <EOS>
5. A
6. C
7. B
8. B
13. B
14. A
15. C
16. A
- token received.